

QuickBooks Subscription Status Validation & Data Synchronization Flow

1. Objective

The purpose of this implementation is to validate the connected QuickBooks Online company's subscription status before performing data synchronization in FinAccrual.

Currently, FinAccrual retrieves QuickBooks data such as Customers and Vendors through QuickBooks APIs and stores the required data in the FinAccrual backend database.

Similarly, we can validate the QuickBooks subscription status through the QuickBooks `CompanyInfo` API.

2. Existing Data Synchronization Flow

The current data flow is:

```
graph TD
    A[QuickBooks] --> B[Customer / Vendor / Account APIs]
    B --> C[FinAccrual Backend]
    C --> D[Validate and process API response]
    D --> E[Store data in FinAccrual Database]
    E --> F[Display data in FinAccrual Frontend]
```

3. QuickBooks Subscription Validation

QuickBooks provides the `SubscriptionStatus` attribute through the `CompanyInfo` entity.

The backend can call the QuickBooks `CompanyInfo` API and check the current `SubscriptionStatus`.

Possible subscription statuses include:

- `TRIAL`
- `TRIALOPTIN`
- `SUBSCRIBED`
- `EXPIRED`
- `RESTRICTED`
- `SUSPENDED`

- CANCELLED
- UNKNOWN

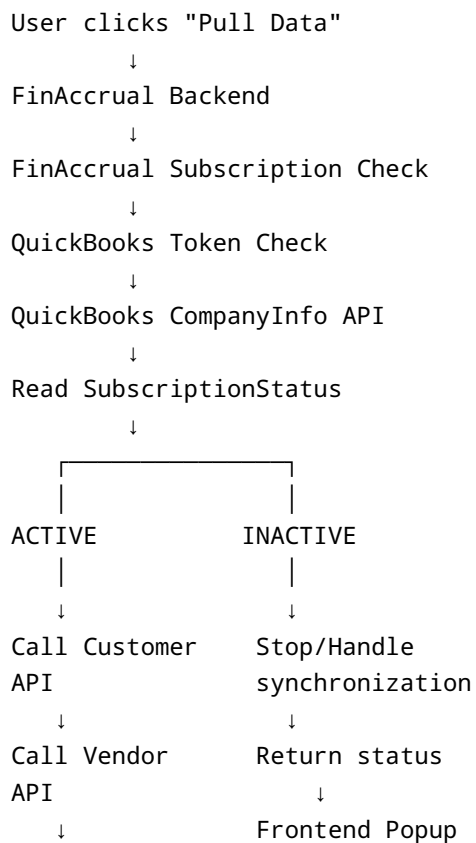
QuickBooks documents that TRIAL and SUBSCRIBED allow read/write operations, while EXPIRED, RESTRICTED, SUSPENDED, and CANCELLED are limited to read-only operations.

4. Proposed FinAccrual Flow

When the user clicks **Pull Data / Full Setup / Sync**, the backend should process the request in the following order:

1. Validate FinAccrual subscription.
2. Validate the QuickBooks OAuth connection/token.
3. Call QuickBooks CompanyInfo API.
4. Read SubscriptionStatus.
5. If the QuickBooks subscription is active, continue with the required data APIs.
6. Retrieve Customers, Vendors, Accounts, and other required data.
7. Process and store the retrieved data in the FinAccrual database.
8. If the QuickBooks subscription is inactive/restricted, stop the normal synchronization flow.
9. Return an appropriate status/error response to the frontend.
10. Display a user-friendly popup/message.

5. Example Flow



```
Call Account
API
↓
Process Data
↓
Store in FinAccrual DB
```

6. Active Subscription

If the QuickBooks response indicates an active subscription, such as `SUBSCRIBED`, the backend continues the normal synchronization process.

Example:

```
CompanyInfo API
↓
SubscriptionStatus = SUBSCRIBED
↓
Continue Sync
↓
Customers API
Vendors API
Accounts API
Other required APIs
↓
Store data in FinAccrual DB
```

7. Expired / Restricted Subscription

If QuickBooks returns an inactive status such as `EXPIRED`, `SUSPENDED`, `RESTRICTED`, or `CANCELLED`, the backend should not continue the normal synchronization flow.

The backend should return a meaningful application-level response to the frontend.

Example frontend message:

"Your QuickBooks subscription is no longer active. Please renew your QuickBooks subscription and try again."

QuickBooks also documents error `6190 - Invalid Company Status` for cases where the subscription period has ended, the subscription was cancelled, or there is a billing-related issue.

8. Database Handling

The current QuickBooks data such as Customers and Vendors can continue to be stored in the FinAccrual database.

For subscription status, the application may store the last-known status and the time it was checked for audit/tracking purposes.

However, the current QuickBooks subscription status should be validated from the QuickBooks API rather than relying only on an old database value.

Example:

```
QuickBooks CompanyInfo API
    ↓
SubscriptionStatus
    ↓
FinAccrual Backend
    ↓
Optional: Store last checked status/time
    ↓
Continue or stop synchronization
```

9. Frontend Behavior

The frontend should not directly determine whether the QuickBooks subscription is active.

The backend should perform the validation and return the result.

Example:

```
Backend Response
    ↓
Frontend
    ↓
If successful
    → Continue / display synced data

If subscription inactive
    → Show subscription-related popup
```

Example popup:

QuickBooks Subscription Inactive

"Your QuickBooks subscription is no longer active. Please renew your QuickBooks subscription and try again."

10. Key Architecture Point

QuickBooks data retrieval and subscription validation are handled through the same QuickBooks integration, but they are separate operations.

Data APIs:

- Customer
- Vendor
- Account
- Invoice
- Other required entities

Subscription validation:

- CompanyInfo
- `SubscriptionStatus`

Therefore, before starting the main synchronization, FinAccrual can validate the QuickBooks company's current subscription status and then decide whether to continue the data synchronization.

11. Expected Result

This implementation provides:

- QuickBooks subscription validation before synchronization.
- Better handling of expired/restricted QuickBooks accounts.
- Prevention of unnecessary data synchronization attempts.
- Clear backend status handling.
- User-friendly frontend popup messages.
- Better separation between QuickBooks subscription status and FinAccrual subscription status.
- A reliable flow for both current development/sandbox testing and future production customer accounts.

Reference

QuickBooks Online official documentation confirms that the `CompanyInfo.SubscriptionStatus` attribute should be used to check an end-user's current QuickBooks Online subscription status.